

Network Layer (3)

COMP90007 Internet Technologies

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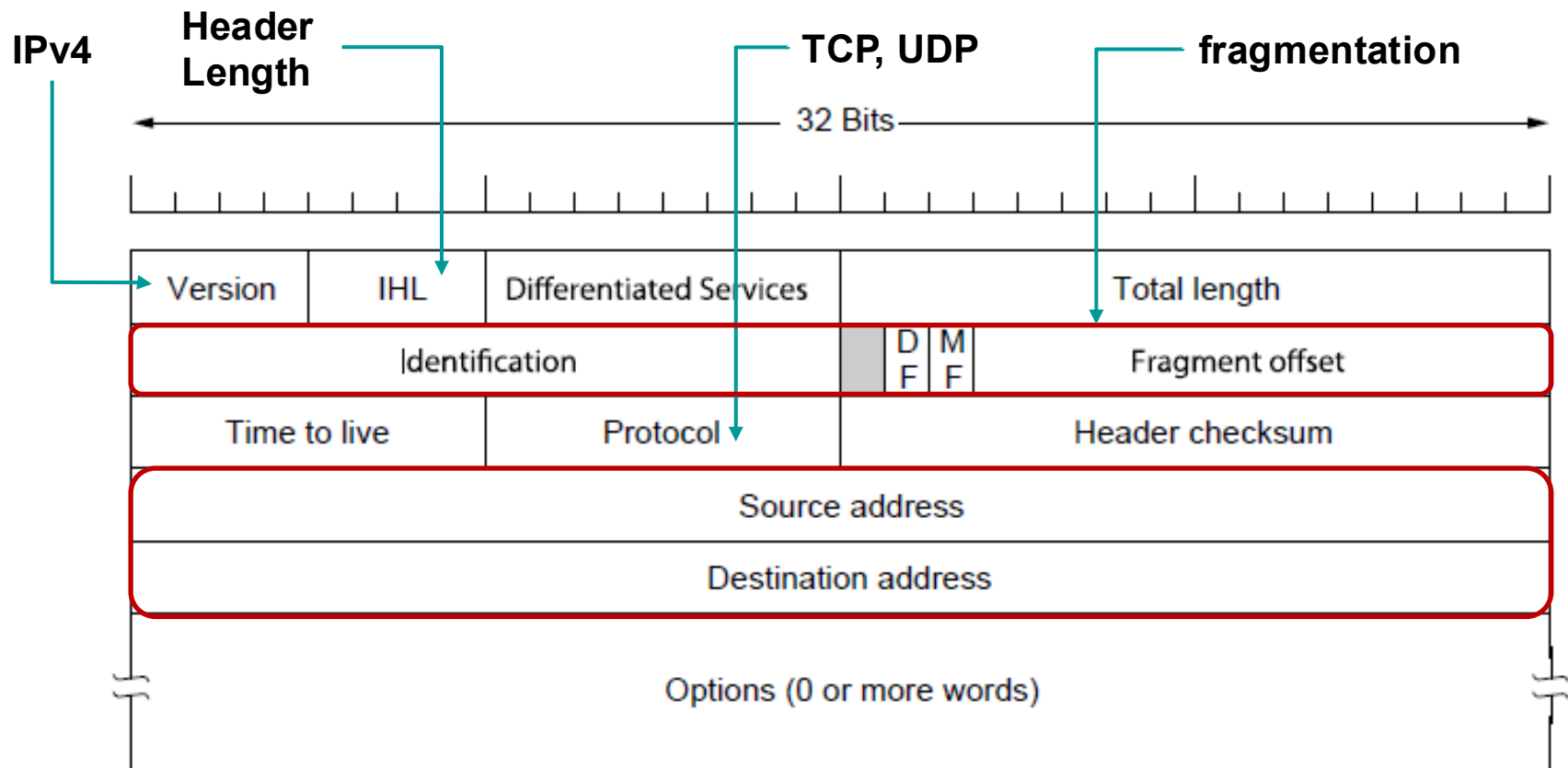
Semester 1, 2026

Outline

- Network layer on the Internet
- Types of services
- Internetworking
 - Tunneling
 - Fragmentation
 - Path MTU discovery
- Internet Protocol
 - Addressing
 - Subnetting
- Routing algorithms

IPv4 Datagram Structure (1)

- **Internet Protocol IPv4 datagram** consists of a header and payload
 - **Header format:** 20-byte fixed part + variable-length optional part



IPv4 Datagram Structure (2)

- **IHL:** Internet Header Length (in 32-bit units), min = 5 and max = 15
- **Differentiated services:** different classes of service
- **Total Length:** max length 65,535 bytes, including header and payload
- **Identification:**
 - Allows the host to determine which datagram the new fragment belongs to
 - All fragments of the same datagram have the same ID
- **DF:** Don't Fragment
 - Used as part of the process for discovering the path MTU
- **MF:** More Fragment. Is this the last one?
- **Fragment offset:** the position of the current fragment within the original datagram

IPv4 Datagram Structure (3)

- **TTL:** Time to Live, the maximum duration a datagram can exist in the network, measured in hops or seconds
- **Protocol:** transport layer protocol, e.g. TCP, UDP
- **Header Checksum:** validates the integrity of the header only

- **Source Address:** IP address of the sender
- **Destination Address:** IP address of the receiver

- **Options:** e.g. security, strict vs. loose source routing, record route, and timestamp

IP Addresses (1)

- IPv4 address is 32-bit long, written in dotted decimal notation

128.18.3.11



2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	1	0	0	1	0

range: 0-255

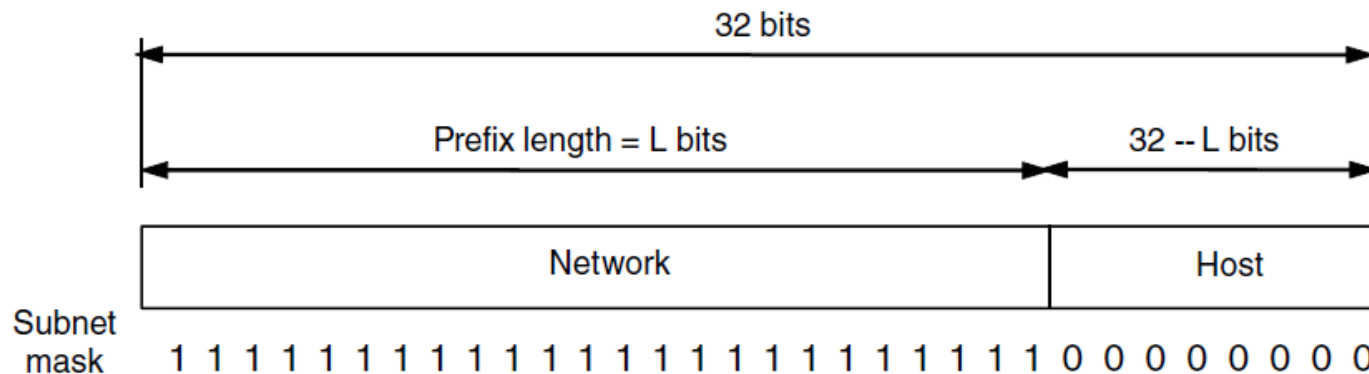
- IP allocation is managed by Internet Corporation for Assigned Names and Numbers (ICANN)
- IPv4 address exhaustion issue?

IP Addresses (2)

- **Hierarchical structure:** network portion + host portion
- **Allocation:** addresses are assigned in **blocks**, all hosts within a block share the same network portion
e.g. a block of 256 addresses 128.18.3.0 - 128.18.3.255
- **Prefix:** identifies an address block, written as
lowest address/bit-length
e.g. 128.18.3.0/24, 18.2.0.0/16

IP Addresses (3)

- **Subnet mask:** all 1s in the network portion
- **Extracting** the network portion: apply a bitwise AND between the IP address and the subnet mask



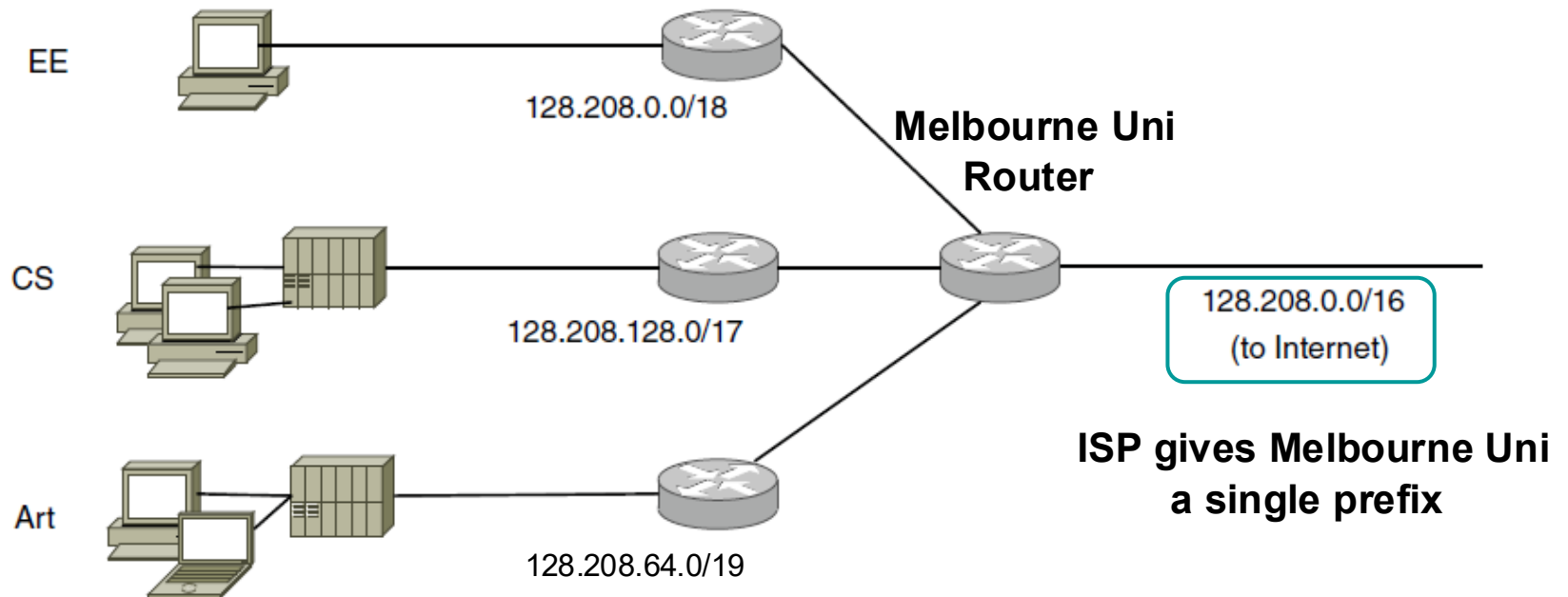
IP Addresses and Routing Tables

- Routing tables are typically built on a triplet:
 - Network destination
 - Subnet mask
 - Outgoing line (physical or virtual)
- Example: a row of a routing table

Destination	Subnet Mask	Interface
128.18.3.0	255.255.255.0	Eth 0

Subnets (1)

- Subnetting allows networks to be split into several parts for internal purposes, while acting like a single network for external users



Example: the network 128.208.0.0/16 is divided into subnets internally, but looks like a single prefix outside the network

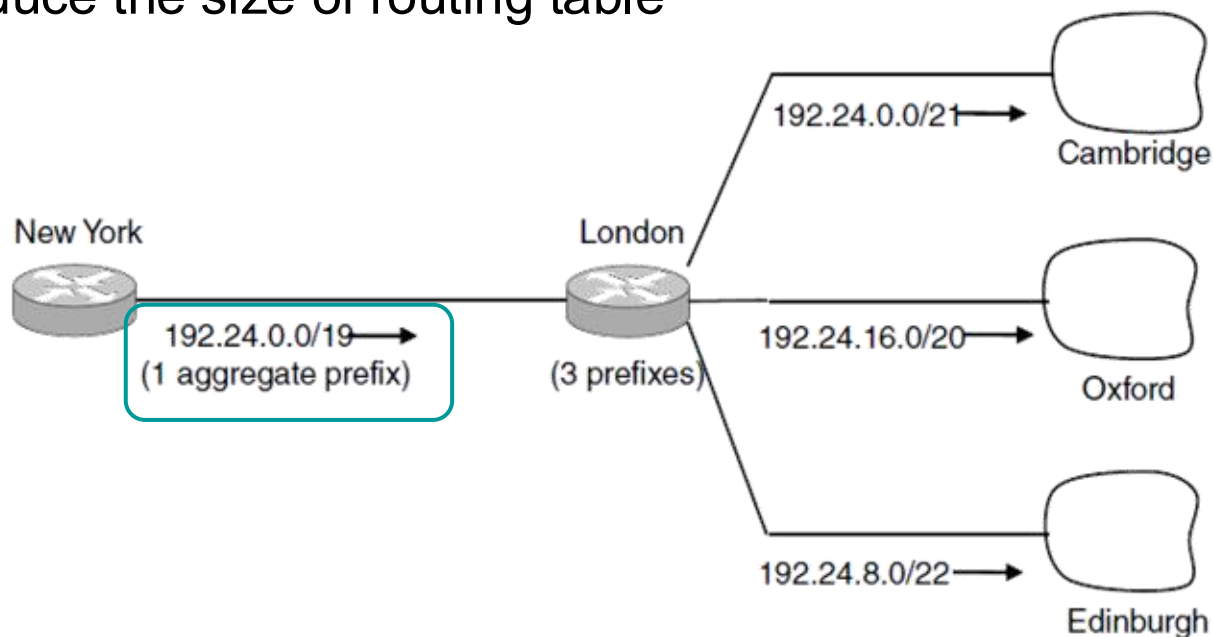
Subnets (2)

128.208.0.0/16 → number of addresses 2^{16}

block	128.208.	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	2^{16}
EE	128.208.	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	2^{14}
Art	128.208.	0	1	0	x	x	x	x	x	x	x	x	x	x	x	x	2^{13}
CS	128.208.	1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	2^{15}
Left	128.208.	0	1	1	x	x	x	x	x	x	x	x	x	x	x	x	2^{13}

Classless Inter-Domain Routing (1)

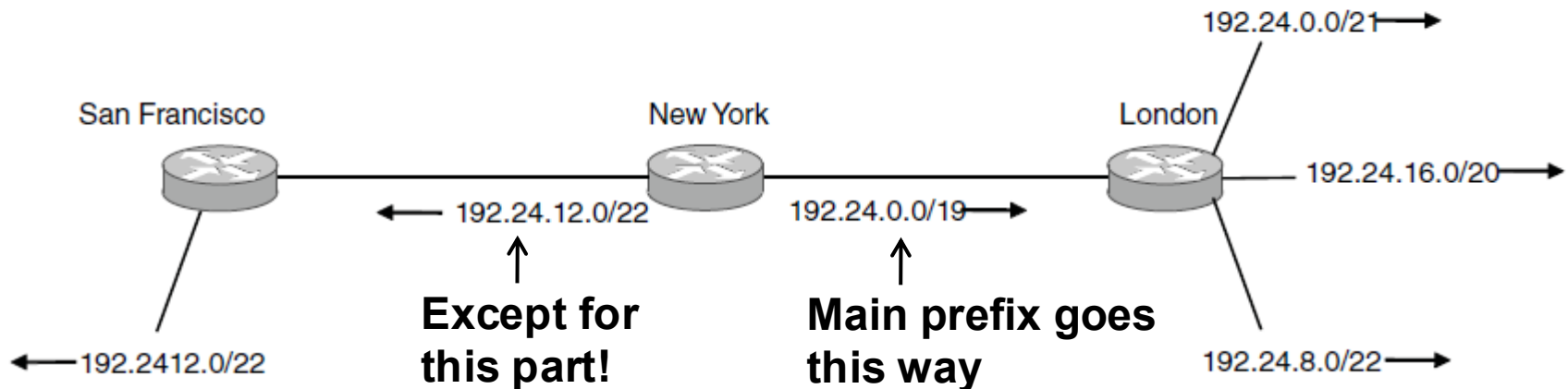
- **Routing table explosion?** Backbone routers connect the networks around the world → 300k networks
- **Aggregation:** joining multiple IP prefixes into a single larger prefix to reduce the size of routing table



Classless Inter-Domain Routing (2)

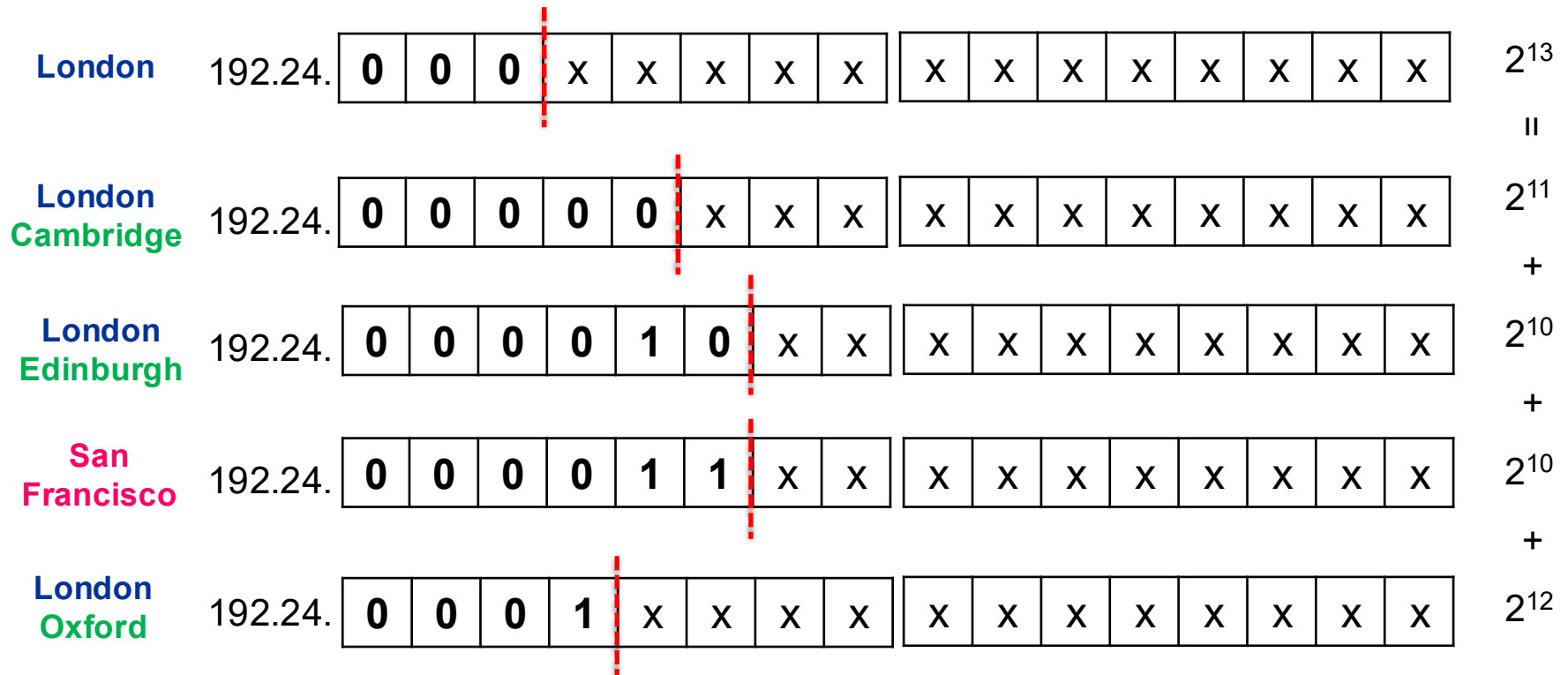
- Packets are forwarded to the entry with the **longest matching prefix** (i.e. the smallest address block)
- Complicates forwarding process but adds flexibility
 - 1) Check whether an address matches the longest prefix **192.24.12.0/22**
 - 2) If not, then check if it matches **192.24.0.0/19**

Destination	Subnet Mask	Interface
192.24.12.0	255.255.252.0	Eth 0 (to SF)
192.24.0.0	255.255.224.0	Eth 1 (to London)



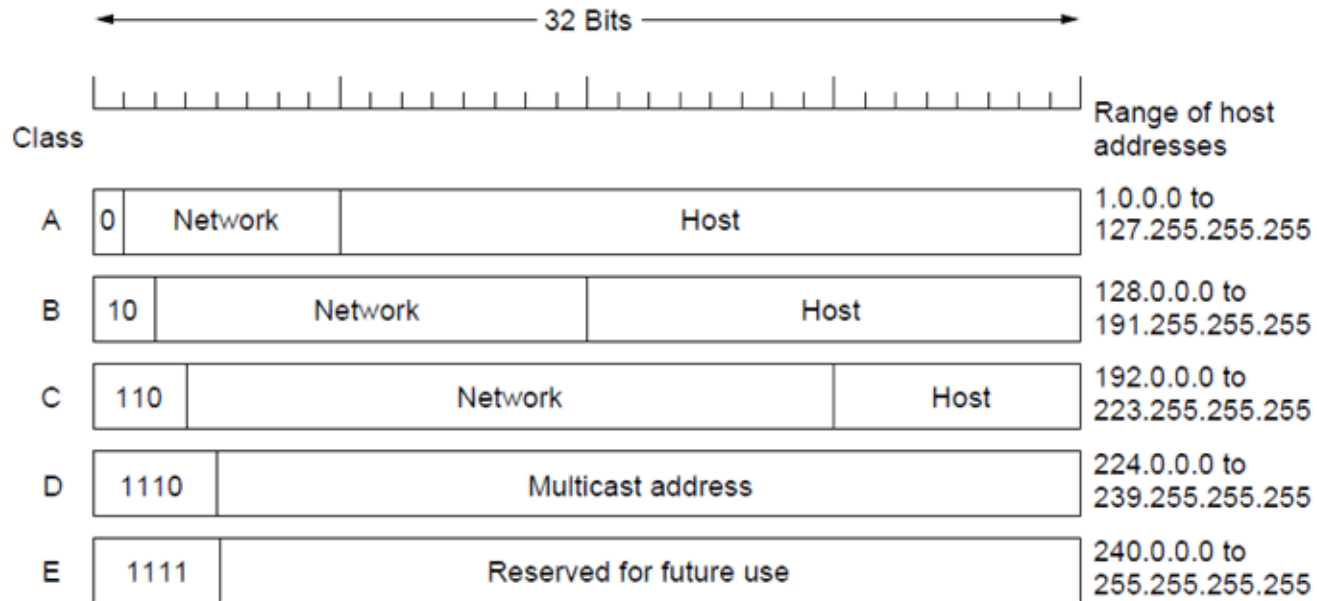
Classless Inter-Domain Routing (3)

192.24.0.0/19 → number of addresses 2^{13} (8192)



Classful Addressing

- Addresses were originally assigned in fixed-size blocks
 - Class A, B, C, D, E
 - Each class encodes its block size within the address, lacks flexibility

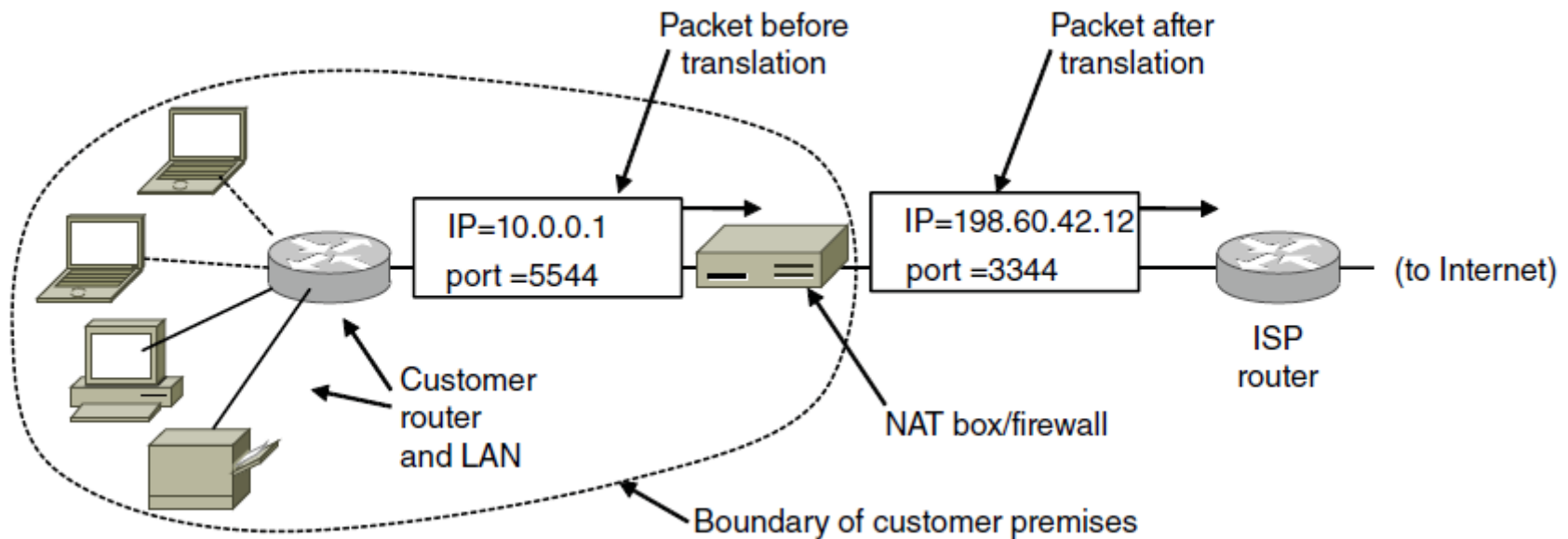


Private IP Ranges

- Range of IP addresses that are NOT publicly routable on the global Internet
- Reserved only for private networks
 - ❑ 10.0.0.0/8 ($2^{24} = 16,777,216$ hosts)
 - ❑ 172.16.0.0/12 ($2^{20} = 1,048,576$ hosts)
 - ❑ 192.168.0.0/16 ($2^{16} = 65,536$ hosts)

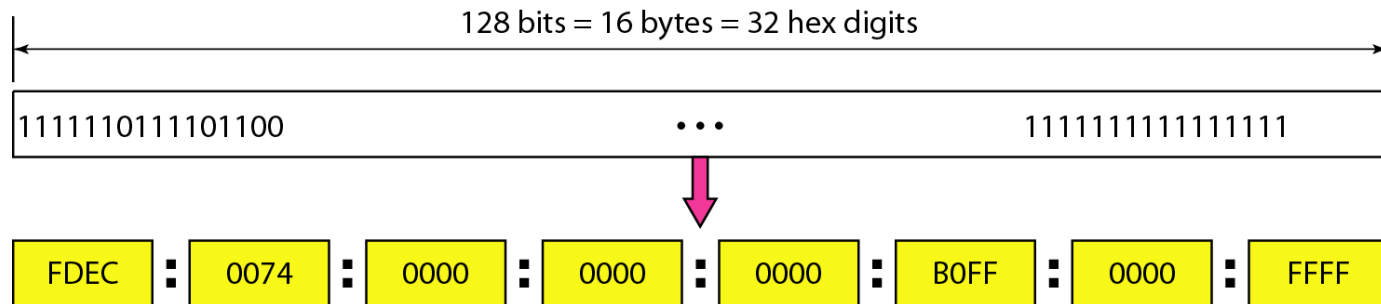
Network Address Translation (NAT)

- NAT box maps one external IP address to multiple internal IP addresses
 - Uses TCP/UDP port numbers to distinguish connections
 - A common strategy for conserving global address space, but it violates the layering principle



IPv6 (1)

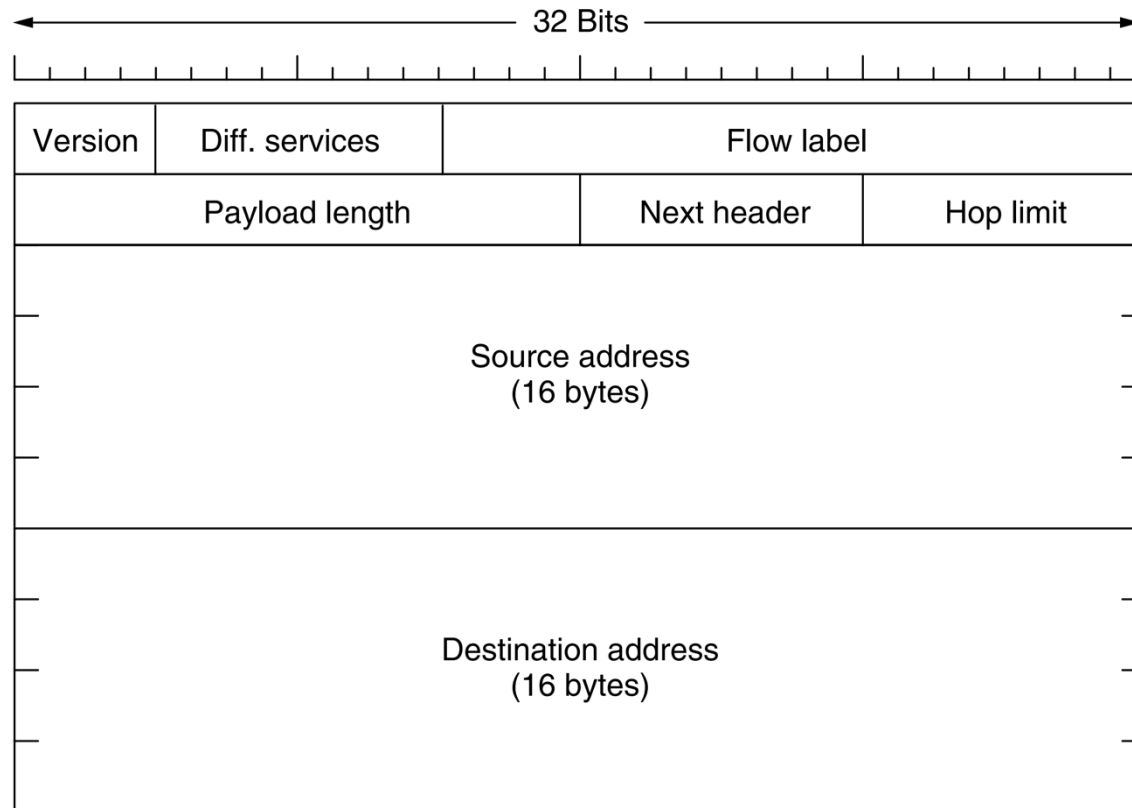
- IPv6 addresses use 128 bits, represented in hexadecimal, separated by colons
 - Significantly larger address space



- Support for more security: encryption and authentication
- Transition: dual stack, tunneling, header translation

IPv6 (2)

- Required fields in IPv6 header (40 bytes)



Internet Control Protocols

- IP works with the help of several control protocols:
 - **ICMP** (Internet Control Message Protocol) is a companion to IP that returns error information
 - e.g. traceroute, ping
 - **ARP** (Address Resolution Protocol) finds the MAC address of a local IP address
 - The host queries an address and the owner replies
 - **DHCP** (Dynamic Host Control Protocol) assigns a local IP address to a host
 - Initiates the host by automatically configuring network settings
 - The host sends a request to the server, which grants a lease for the IP address

ICMP

- Used for testing and monitoring ambient conditions between hosts and routers

	Message type	Description
➡	Destination unreachable	Packet could not be delivered
➡	Time exceeded	Time to live field hit 0
	Parameter problem	Invalid header field
	Source quench	Choke packet
	Redirect	Teach a router about geography
➡	Echo and Echo reply	Check if a machine is alive
	Timestamp request/reply	Same as Echo, but with timestamp
	Router advertisement/solicitation	Find a nearby router